



## Intraspecific variability (rDNA ITS and mtDNA Cyt *b*) of *Phlebotomus sergenti* in Spain and Morocco

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### ABSTRACT

*Phlebotomus sergenti*, the main vector of *Leishmania tropica*, is widely represented in Spain, whilst *L. tropica* is not an endemic species in this country. Nevertheless, the important human migrant flow from regions where *L. tropica* is endemic, the existence of its vector and the anthroponotic cycle of the parasite could lead to its establishment in our country. The vectorial capacity of *P. sergenti* could depend on the existence of cryptic species or races, which can only be identified by molecular methods. Our aim has been to study the molecular variation on two genes of Spanish populations of *P. sergenti*, comparing them with Moroccan ones. We have identified 5 ribosomal and 16 mitochondrial haplotypes. The results obtained indicate a high diversity of *P. sergenti* in Spain and the existence in the country of two *P. sergenti* lineages, a typically Spanish mitochondrial lineage and another one common in Morocco, where *L. tropica* is endemic in the south and emergent in the north and centre of the country.

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### 1. Introduction

*Phlebotomus* (*Paraphlebotomus*) *sergenti* Parrot, 1917 is the main vector of *Leishmania tropica* (Wright, 1903), responsible for the anthroponotic cutaneous leishmaniasis (ACL) (Al-Zahrani et al., 1988; Guilvard et al., 1991). It has an extensive geographical distribution, wider than that of the flagellate (Depaquit et al., 2002). It has been suggested that the presence of this sandfly in *L. tropica* free areas and the differences in the transmission patterns of the ACL could be related to the existence of races or even cryptic vector species (Depaquit et al., 2002; Yahia et al., 2004). *Phlebotomus sergenti* is a species widely present in Spain (Gil Collado et al., 1989). In Spain it shows a biphasic phenology in localities with optimal climatology for its development, with two maxima located at the beginning and at the end of the summer period, that are separated by a minimum in the month of August (Martínez Ortega and Conesa Gallego, 1987; Sanchís-Marín et al., 1986), similar to the phenology of this species in Morocco (Guernaoui et al., 2005; Guessous-Idrissi et al., 1997). Conversely, *L. tropica* is not an endemic species in Spain, where only *Leishmania infantum* has this

trait (Gállego et al., 2002; Martín-Sánchez et al., 2004). Nevertheless (i) the important human migratory flow from areas where this parasitosis is endemic, (ii) the existence of the vector and (iii) the anthroponotic cycle of the parasite, could lead to its establishment in Spain, in a similar way to what has happened in the north of Morocco (Guessous-Idrissi et al., 1997; Rhajaoui et al., 2004). Bearing in mind that the vectorial capacity of *P. sergenti* could depend on the existence of races or even sibling species, which can be identified only by molecular methods, our aim has been to study the molecular variation of Spanish *P. sergenti* populations, comparing them with Moroccan ones. For that reason, we have selected two genes with a high mutation rate and a different type of heredity, such as are the internal transcribed spacer 2 (ITS2) of the ribosomal DNA (rDNA) and a fragment of the gene Cytochrome *b* (Cyt *b*) of the mitochondrial DNA (mtDNA). In both cases the scientific literature describes primers and conditions for the PCR and with pre-existing knowledge of them in some sandfly species. The ITS2 has provided resolution in several studies at the subgeneric taxonomic level for the *Larroussius* and *Paraphlebotomus* (Depaquit et al., 2000, 2002; Di Muccio et al., 2000). The comparative sequence analysis of mtDNA fragments has been shown to be useful for genetic differentiation of geographical lineages and species on the subgenus *Larroussius*, *Phlebotomus* and *Paraphlebotomus* (Esseghir et al., 1997; Moin-Vaziri et al., 2007; Perrotey et al., 2005; Pesson et al., 2004; Yahia et al., 2004).

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## 2. Material and methods

### 2.1. Study areas

The geographical locations from which *P. sergenti* was sampled are shown in Fig. 1. Latitude and longitude, of the different sampled localities, are indicated.

### 2.2. Collection and morphological identification of specimens

Sandflies were captured overnight with miniature CDC light traps and preserved in liquid nitrogen or at  $-20^{\circ}\text{C}$  until morphological and molecular identification (Table 1). They were slide-mounted in Berlese fluid following dissection and morphologically identified based on external and internal characters of the head and genitalia according to the keys of Gil Collado et al. (1989) and Gállego et al. (1992).

### 2.3. Sandfly DNA extraction

Genomic DNA was extracted from the thorax and the attached anterior abdomen of individual sandflies (Martín-Sánchez et al., 2000). Briefly, each sandfly was placed in a 1.5-ml Eppendorf tube and kept in liquid nitrogen for a few seconds to facilitate the rupture of cell membranes. The tissue was crushed by a pestle in lysis buffer (10  $\mu\text{l}$  of SCE (sorbitol 1 M, sodium citrate 0.1 M, EDTA 0.06 M), 30  $\mu\text{l}$  of SDS-EDTA (SDS 1%, EDTA 0.15 M) and 10  $\mu\text{l}$  of proteinase K 1%) and then incubated at  $42^{\circ}\text{C}$  overnight. Subsequently, the DNA was purified with one phenol–chloroform–isoamyl alcohol extraction, followed by one chloroform–isoamyl alcohol extrac-

tion and then an ethanol precipitation. The precipitated pellet was resuspended in 50  $\mu\text{l}$  of bidistilled water and kept at  $-20^{\circ}\text{C}$  until use.

### 2.4. PCR amplification

The polymerase chain reaction was used to amplify a fragment of 418–433 bp containing the ITS2 of sandfly rDNA following the methodology described by Depaquit et al. (2002).

PCR was also used to amplify a 550-bp fragment containing the 3' end of the mitochondrial gene Cyt *b* following the methodology described by Essegir et al. (1997). In the specimens in which this PCR did not work, an internal fragment of 442 bp was amplified following the methodology described by Yahia et al. (2004).

### 2.5. Sequencing and comparative sequence analysis

PCR products were directly sequenced in both directions using the primers used for DNA amplification. Sequences were edited and aligned to identify haplotypes (=unique sequences) using Clustal-X 1.81 software and manually adjusted, if necessary. Poly(AT) was excluded from ITS sequences. Phylogenetic analysis was performed using PHYLIP version 3.65 (<http://evolution.genetics.washington.edu/phylip>). *Phlebotomus longicuspis* was used as an outgroup.

## 3. Results

Comparative ITS2 sequence analysis of *P. sergenti* (Table 2): we have analysed this rDNA sequence in 25 *P. sergenti* individuals



**Fig. 1.** Geographical locations of the *P. sergenti* sampled populations with indication of the co-ordinates: (1) Sant Jaume de Llierca, Girona province ( $42^{\circ}12'\text{N}$ ,  $2^{\circ}37'\text{E}$ ), (2) Sant Just Desvern, Barcelona province ( $41^{\circ}22'\text{N}$ ,  $2^{\circ}04'\text{E}$ ), (3) Puerto de la Cadena, Murcia province ( $37^{\circ}09'\text{N}$ ,  $1^{\circ}13'\text{W}$ ), (4) Torvizcón, Granada province ( $36^{\circ}53'\text{N}$ ,  $3^{\circ}18'\text{W}$ ), (5) Almáchar, Málaga province ( $36^{\circ}48'\text{N}$ ,  $4^{\circ}13'\text{W}$ ), (6) Benaolán, Málaga province ( $36^{\circ}43'\text{N}$ ,  $5^{\circ}15'\text{W}$ ), (7) Las Caballerizas, Málaga province ( $37^{\circ}02'\text{N}$ ,  $4^{\circ}43'\text{W}$ ), (8) Riotinto, Huelva province ( $37^{\circ}41'\text{N}$ ,  $6^{\circ}33'\text{W}$ ), (9) Chechaouene, Chaouen province ( $34^{\circ}53'\text{N}$ ,  $3^{\circ}49'\text{W}$ ), (10) Taounate, Taounate province ( $34^{\circ}32'\text{N}$ ,  $7^{\circ}59'\text{W}$ ) and (11) Dar Bachir, Marrakech province ( $31^{\circ}31'\text{N}$ ,  $7^{\circ}28'\text{W}$ ).

**Table 1**  
List of the individuals used in this study

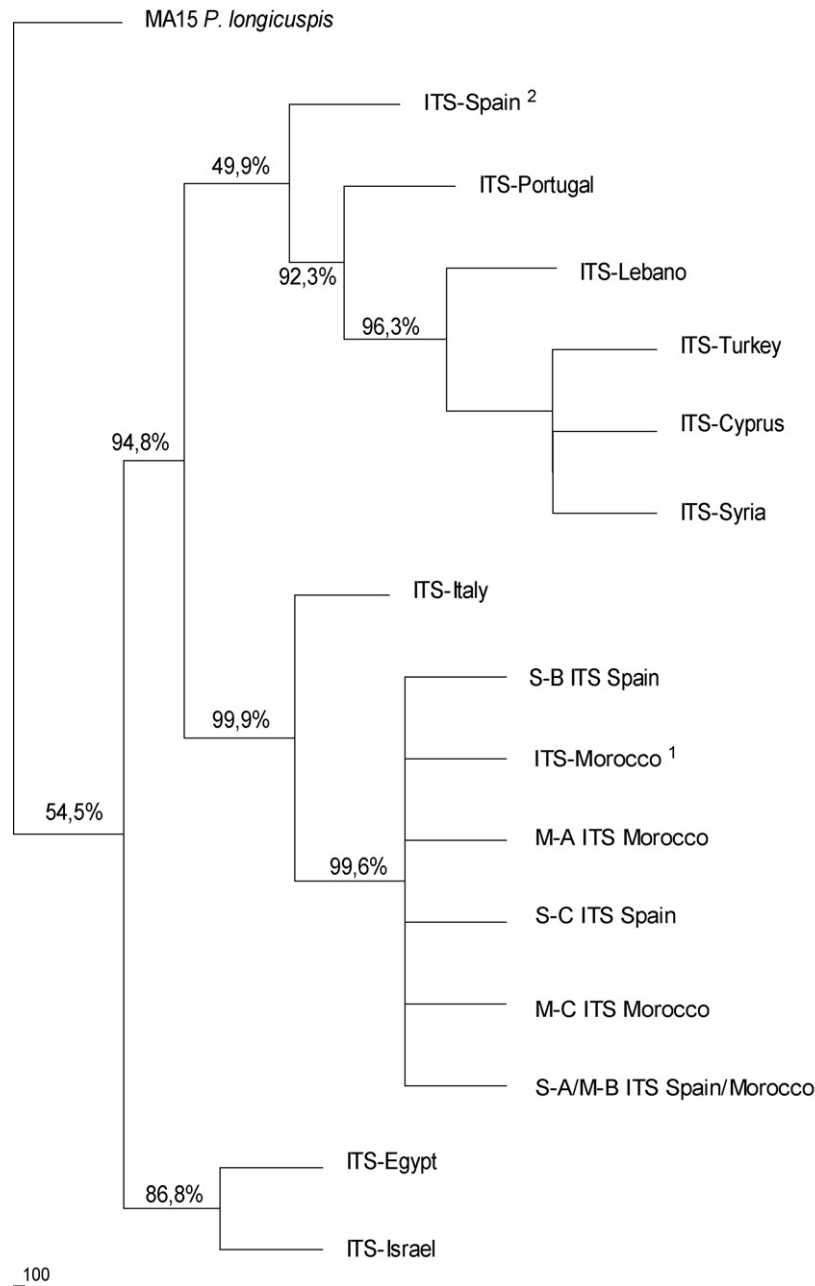
| Specimen code | Sex: M or F | Origin (country, locality and number assigned in Fig. 1) and year of catching | Studied markers      | Gen Bank accession number |          |
|---------------|-------------|-------------------------------------------------------------------------------|----------------------|---------------------------|----------|
|               |             |                                                                               |                      | Cyt <i>b</i>              | ITS      |
| P-6           | M           | Spain, Benaolan, 6, 2003                                                      | Cyt <i>b</i>         | EU980375                  |          |
| P-7           | M           | Spain, Benaolan, 6, 2003                                                      | Cyt <i>b</i>         | EU980375                  |          |
| P-17          | M           | Spain, Las Caballerizas, 7, 2003                                              | ITS and Cyt <i>b</i> | EU980375                  | EU980385 |
| P-46          | M           | Spain, Las Caballerizas, 7, 2003                                              | ITS and Cyt <i>b</i> | EU980375                  | EU980386 |
| P-47          | M           | Spain, Las Caballerizas, 7, 2003                                              | Cyt <i>b</i>         | EU980375                  | -        |
| P-48          | M           | Spain, Las Caballerizas, 7, 2003                                              | Cyt <i>b</i>         | EU980375                  | -        |
| P-49          | F           | Spain, Las Caballerizas, 7, 2003                                              | Cyt <i>b</i>         | EU980378                  | -        |
| P-50          | F           | Spain, Las Caballerizas, 7, 2003                                              | ITS and Cyt <i>b</i> | EU980376                  | EU980385 |
| P-52          | F           | Spain, Las Caballerizas, 7, 2003                                              | ITS and Cyt <i>b</i> | EU980378                  | EU980385 |
| AM-1          | M           | Spain, Almachar, 5, 1999                                                      | Cyt <i>b</i>         | EU980375                  | -        |
| AM-2          | F           | Spain, Almachar, 5, 1999                                                      | ITS and Cyt <i>b</i> | EU980375                  | EU980385 |
| AM-3          | F           | Spain, Almachar, 5, 1999                                                      | ITS and Cyt <i>b</i> | EU980380                  | EU980385 |
| AM-4          | M           | Spain, Almachar, 5, 1999                                                      | Cyt <i>b</i>         | EU980381                  | -        |
| AM-5          | M           | Spain, Almachar, 5, 1999                                                      | Cyt <i>b</i>         | EU980380                  | -        |
| AM-6          | F           | Spain, Almachar, 5, 1999                                                      | ITS                  | -                         | EU980385 |
| TM-1          | M           | Spain, Torvizcón, 4, 1999                                                     | ITS and Cyt <i>b</i> | EU980380                  | EU980385 |
| TM-2          | F           | Spain, Torvizcón, 4, 1999                                                     | Cyt <i>b</i>         | EU980380                  | -        |
| TM-3          | F           | Spain, Torvizcón, 4, 1999                                                     | ITS and Cyt <i>b</i> | EU980380                  | EU980385 |
| TM-4          | M           | Spain, Torvizcón, 4, 1999                                                     | Cyt <i>b</i>         | EU980380                  | -        |
| TM-5          | F           | Spain, Torvizcón, 4, 1999                                                     | Cyt <i>b</i>         | EU980375                  | -        |
| TM-6          | M           | Spain, Torvizcón, 4, 1999                                                     | Cyt <i>b</i>         | EU980375                  | -        |
| TM-7          | F           | Spain, Torvizcón, 4, 1999                                                     | Cyt <i>b</i>         | EU980380                  | -        |
| TM-8          | M           | Spain, Torvizcón, 4, 1999                                                     | ITS and Cyt <i>b</i> | EU980375                  | EU980385 |
| TM-9          | M           | Spain, Torvizcón, 4, 1999                                                     | Cyt <i>b</i>         | EU980375                  | -        |
| P-121         | M           | Spain, Torvizcón, 4, 2002                                                     | Cyt <i>b</i>         | EU980380                  | -        |
| P-124         | M           | Spain, Torvizcón, 4, 2002                                                     | Cyt <i>b</i>         | EU980380                  | -        |
| P-127         | M           | Spain, Torvizcón, 4, 2002                                                     | Cyt <i>b</i>         | EU980380                  | -        |
| P-128         | M           | Spain, Torvizcón, 4, 2002                                                     | Cyt <i>b</i>         | EU980380                  | -        |
| P-131         | F           | Spain, Torvizcón, 4, 2002                                                     | Cyt <i>b</i>         | EU980380                  | -        |
| R-9           | M           | Spain, P. de la Cadena, Murcia, 3, 1999                                       | ITS                  | -                         | EU980387 |
| P-137         | F           | Spain, Sant Just Desvern, 2, 2002                                             | ITS and Cyt <i>b</i> | EU980375                  | EU980385 |
| P-139         | M           | Spain, Sant Just Desvern, 2, 2002                                             | ITS                  | -                         | EU980385 |
| P-141         | M           | Spain, Sant Just Desvern, 2, 2002                                             | Cyt <i>b</i>         | EU980380                  | -        |
| P-143         | F           | Spain, Sant Just Desvern, 2, 2002                                             | ITS and Cyt <i>b</i> | EU980379                  | EU980385 |
| P-146         | M           | Spain, Sant Just Desvern, 2, 2002                                             | Cyt <i>b</i>         | EU980380                  | -        |
| P-149         | M           | Spain, Sant Just Desvern, 2, 2002                                             | ITS                  | -                         | EU980385 |
| P-156         | M           | Spain, Sant Just Desvern, 2, 2002                                             | Cyt <i>b</i>         | EU980380                  | -        |
| P-184         | F           | Spain, Sant Jaume de Llierca, 1, 2002                                         | ITS and Cyt <i>b</i> | EU980375                  | EU980385 |
| RT-1          | M           | Spain, Riotinto, 8, 2000                                                      | Cyt <i>b</i>         | EU980377                  | -        |
| RT-2          | F           | Spain, Riotinto, 8, 2000                                                      | ITS and Cyt <i>b</i> | EU980375                  | EU980385 |
| CH-14         | M           | Morocco, Chechaouene, 9, 1995                                                 | Cyt <i>b</i>         | EU980380                  | -        |
| MA-1          | F           | Morocco, Taounate, 10, 1995                                                   | ITS and Cyt <i>b</i> | EU980364                  | EU980383 |
| MA-2          | M           | Morocco, Taounate, 10, 1995                                                   | ITS and Cyt <i>b</i> | EU980366                  | EU980383 |
| MA-3          | M           | Morocco, Taounate, 10, 1995                                                   | Cyt <i>b</i>         | EU980365                  | -        |
| MA-4          | M           | Morocco, Taounate, 10, 1995                                                   | Cyt <i>b</i>         | EU980365                  | -        |
| MA-5          | F           | Morocco, Taounate, 10, 1995                                                   | Cyt <i>b</i>         | EU980367                  | -        |
| MA-7          | M           | Morocco, Taounate, 10, 2005                                                   | Cyt <i>b</i>         | EU980368                  | -        |
| MA-8          | M           | Morocco, Taounate, 10, 1995                                                   | Cyt <i>b</i>         | EU980364                  | -        |
| MA-9          | F           | Morocco, Taounate, 10, 2005                                                   | ITS and Cyt <i>b</i> | EU980368                  | EU980383 |
| MA-10         | M           | Morocco, Taounate, 10, 1995                                                   | Cyt <i>b</i>         | EU980364                  | -        |
| MA-11         | M           | Morocco, Taounate, 10, 1995                                                   | Cyt <i>b</i>         | EU980368                  | -        |
| MA-14         | M           | Morocco, Dar Bachir, 11, 2005                                                 | ITS and Cyt <i>b</i> | EU980370                  | EU980382 |
| MA-16         | M           | Morocco, Dar Bachir, 11, 2005                                                 | ITS and Cyt <i>b</i> | EU980370                  | EU980382 |
| MA-17         | F           | Morocco, Dar Bachir, 11, 2005                                                 | ITS and Cyt <i>b</i> | EU980369                  | EU980382 |
| MA-18         | M           | Morocco, Dar Bachir, 11, 2005                                                 | ITS and Cyt <i>b</i> | EU980373                  | EU980382 |
| MA-19         | F           | Morocco, Dar Bachir, 11, 2005                                                 | Cyt <i>b</i>         | EU980370                  | -        |
| MA-20         | M           | Morocco, Dar Bachir, 11, 2005                                                 | ITS and Cyt <i>b</i> | EU980371                  | EU980384 |
| MA-21         | M           | Morocco, Dar Bachir, 11, 2005                                                 | Cyt <i>b</i>         | EU980372                  | -        |

collected in 7 Spanish localities (17 specimens) and 2 Moroccan ones (8 specimens). Five different haplotypes have been identified: 2 in the Spanish populations exclusively (S-B ITS and S-C ITS), another 2 in the Moroccan ones (M-A ITS and M-C ITS) and one (S-A/M-B ITS) shared by both of them. These haplotypes differ in only 1–3 nucleotides. S-A/M-B ITS is the ribosomal haplotype most frequently and widely represented among the Spanish populations of *P. sergenti* being found in all the localities sampled except in Murcia (locality 6). It is also the only haplotype that

we found in the Rif (north of Morocco) although it is absent in the south. Sequences were identical for specimens sampled from the same location except in Las Caballerizas (locality 7, south of Spain) with two haplotypes and in Dar Bachir (locality 11, south of Morocco) with another two, one of them was heterozygote, in three sites.

Parsimony analysis identified a single tree of 159 steps. The topology was similar to that shown by the tree based on distance analysis (F84 model with a UPGMA method of clustering). None of

[illegible]



**Fig. 2.** rDNA ITS2 phylogenetic tree including the sequences obtained by Depaquit et al. (2002). The tree is generated by maximum parsimony method. *Phlebotomus longicuspis* is used as outgroup. The numbers above the branches are bootstrap percentages (1000 replications) for clades supported above the 50%. <sup>1</sup>Its sequence is coincident with the S-A/M-B ITS sequence. <sup>2</sup>This sequence shows differences in 13–16 nucleotides to the Spanish ones sequenced by us. The ecological niche of this population was destroyed as a consequence of human activities shortly after its capture in 1994. Hence we have not been able to incorporate this population into our study.

we have kept in mind the previously published sequences (Yahia et al., 2004).

**Combined analysis of ITS2 and Cyt *b* sequences:** We have analysed both sequences in 12 *P. sergenti* individuals collected in 6 Spanish localities and 8 specimens collected in 2 Moroccan ones. Table 5 shows the haplotypes of both markers which are found associated in each one of these individuals.

#### 4. Discussion

The obtained results show a high diversity between specimens of *P. sergenti* in Spain where we find 3 ribosomal and 7 mitochon-

drial haplotypes. Some of the genetic variants are also present in Morocco where the polymorphism of *P. sergenti* is larger (3 ribosomal and 10 mitochondrial haplotypes). The 5 ITS haplotypes identified in this study are closely related to each other and belong to the same ribosomal lineage (Fig. 2). The information given by this marker is much smaller than the one observed in the mitochondrial sequences that seem to be more appropriate for the study of the intraspecific variability of *P. sergenti* in a more limited geographical environment. Thus, the 16 Cyt *b* haplotypes are distributed in 4 mitochondrial lineages (which we have named I, II, III, and IV lineages) well differentiated and well supported in the bootstrap analysis (Fig. 3a). The mitochondrial lineage I corresponds



**Table 4**  
Alignment of nucleotide characters at the polymorphic sites in the mt DNA Cytochrome *b* haplotypes of *P. sergenti* with indication of the synapomorphic (highlighted) and the variable characters (underlined) for each lineage

| LINEAGE | Cyt b<br>Haplotypes |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---------|---------------------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
|         |                     | 1 | 3 | 6 | 1 | 2 | 3 | 4 | 4 | 4 | 5 | 6 | 6 | 6 | 6 | 7 | 8 | 9 | 9 | 0 | 0 | 1 | 1 | 2 | 4 | 4 | 4 | 5 | 6 | 6 | 7 | 7 | 8 | 8 | 9 | 1 | 2 | 2 | 2 | 2 | 3 | 3 |
| IV      | M-J                 | A | T | T | T | C | A | C | C | T | T | C | A | G | T | T | C | A | T | C | A | T | T | G | C | A | G | T | C | T | T | T | T | C | C | T | C | T | C | T | G | A |
| II      | M-G                 | A | C | C | T | C | A | C | T | T | C | C | G | G | T | T | T | A | T | C | A | C | C | A | C | A | G | C | T | A | C | C | C | T | C | T | C | T | G | A |   |   |
|         | M-H                 | A | C | C | T | C | A | T | T | C | C | G | G | T | T | T | A | T | C | A | C | C | A | C | A | G | C | T | A | C | C | C | T | C | T | C | T | G | A |   |   |   |
|         | M-I                 | A | C | C | T | C | A | C | T | T | C | C | G | G | T | T | T | A | T | C | A | C | C | A | T | A | G | C | T | A | C | C | C | T | C | T | C | T | G | A |   |   |
| III     | M-A / S-F           | A | C | C | T | C | T | C | C | T | T | A | A | C | C | T | G | T | C | A | C | C | A | C | A | A | G | T | T | C | C | T | C | T | C | A | T | T | A | A |   |   |
|         | M-B                 | A | C | C | T | C | T | C | C | T | T | A | A | C | C | T | G | T | T | A | C | C | A | C | A | G | T | T | C | C | T | C | T | C | A | T | T | A | A |   |   |   |
|         | M-C                 | A | C | C | T | T | T | C | C | T | T | A | A | C | C | T | G | T | T | A | C | C | G | C | A | G | T | T | C | C | T | C | T | C | A | T | T | A | A |   |   |   |
|         | M-D                 | A | C | C | C | C | T | C | C | T | T | A | A | C | C | T | G | T | C | A | C | C | A | C | A | G | T | T | C | C | T | C | T | C | A | T | T | A | A |   |   |   |
|         | M-E                 | A | C | C | T | C | T | C | C | T | T | A | A | C | C | T | G | T | C | A | C | C | A | C | T | G | T | T | C | C | T | C | T | C | A | T | T | A | A |   |   |   |
|         | M-F                 | A | C | T | T | C | T | C | C | T | T | A | A | C | C | T | A | T | C | G | C | C | A | C | A | A | T | T | C | C | T | C | T | C | A | T | T | A | A |   |   |   |
|         | S-G                 | A | C | C | T | C | T | C | C | T | T | A | A | C | C | T | G | T | C | A | C | C | A | C | A | A | T | T | C | C | T | C | T | C | A | T | T | A | A |   |   |   |
| I       | S-A                 | G | C | C | T | C | A | C | C | C | T | C | G | G | T | T | T | G | G | C | A | C | C | G | C | A | G | C | T | C | C | T | C | T | C | C | C | T | A | T |   |   |
|         | S-B                 | G | C | C | T | C | A | C | C | C | T | C | G | G | T | T | T | G | G | C | A | C | C | G | C | A | G | C | T | C | C | T | C | T | T | C | C | T | A | T |   |   |
|         | S-C                 | A | C | C | T | C | A | C | C | C | T | C | G | G | T | T | T | G | G | C | A | C | C | G | C | A | G | C | T | C | C | T | C | T | C | C | C | T | A | T |   |   |
|         | S-D                 | G | C | C | T | C | A | C | C | C | T | C | G | G | T | T | T | G | G | C | A | C | C | G | C | A | G | C | T | C | C | T | C | T | C | C | C | C | A | T |   |   |
|         | S-E                 | G | C | C | T | C | A | C | C | C | T | C | G | G | T | T | T | G | G | C | A | C | C | G | C | A | G | C | T | C | C | T | C | T | C | C | T | A | T |   |   |   |

to the Spanish lineage of *P. sergenti* with S-A Cyt *b* being the most representative haplotype which shows a wider geographical distribution. The mitochondrial lineage III is represented in Spain mainly by the haplotype M-A/S-F Cyt *b* and in a lesser degree by S-G Cyt *b*, closely related with the former and from which it may have derived by mutation. This lineage is widely represented in the north of Morocco where it shows a larger genetic diversity: 6 haplotypes differing from 1 to 5 nucleotides. The mitochondrial lineages II and IV, integrated by 3 and 1 haplotypes, respectively, are present only in the south of Morocco.

The larger genetic diversity of *P. sergenti* in Morocco is consistent with the hypothesis of dispersion of *P. sergenti* expressed by

Depaquit et al. (2000, 2002). According to these authors, *P. sergenti* would have colonised Spain from Morocco, before the opening of the Strait of Gibraltar, approximately 5 million years ago. The clustering in the rDNA ITS2 phylogenetic tree (Fig. 2) of the populations from Spain, Morocco and Italy is in agreement with the migratory route proposed for this Phlebotomine sandfly by these authors. Presumably, the Spanish variants would have originated later on by mutation from these early settlers.

M-A/S-F Cyt *b* represents 44.4% of the whole of the Spanish sequenced samples, against the 36.1% represented by S-A Cyt *b*, the representative haplotype of the Spanish mitochondrial lineage. However, M-A/S-F Cyt *b* has a more limited geographi-

**Table 5**  
Relation of individuals in which it has been possible to sequence both markers with indication of the haplotypes ITS and Cyt *b* found associated

| Specimen code | ITS           | Cyt <i>b</i>           |            | Geographical location (number in Fig. 1) |
|---------------|---------------|------------------------|------------|------------------------------------------|
|               | ITS haplotype | Cyt <i>b</i> haplotype | mt lineage |                                          |
| P-17          | S-A/M-B       | S-A                    | I          | Las Caballerizas (7)                     |
| P-46          | S-B           | S-A                    | I          | Las Caballerizas                         |
| P-50          | S-A/M-B       | S-C                    | I          | Las Caballerizas                         |
| P-52          | S-A/M-B       | S-B                    | I          | Las Caballerizas                         |
| AM-2          | S-A/M-B       | S-A                    | I          | Almáchar (5)                             |
| AM-3          | S-A/M-B       | M-A/S-F                | III        | Almáchar                                 |
| TM-1          | S-A/M-B       | M-A/S-F                | III        | Torvizcón (4)                            |
| TM-3          | S-A/M-B       | M-A/S-F                | III        | Torvizcón                                |
| TM-8          | S-A/M-B       | S-A                    | I          | Torvizcón                                |
| P-137         | S-A/M-B       | S-A                    | I          | Sant Just Desvern (2)                    |
| P-143         | S-A/M-B       | S-D                    | I          | Sant Just Desvern                        |
| P-184         | S-A/M-B       | S-A                    | I          | Sant Jaume de Llierca (1)                |
| RT-2          | S-A/M-B       | S-A                    | I          | Riotinto (8)                             |
| MA-1          | S-A/M-B       | M-A/S-F                | III        | Taounate (10)                            |
| MA-2          | S-A/M-B       | M-C                    | III        | Taounate                                 |
| MA-9          | S-A/M-B       | M-E                    | III        | Taounate                                 |
| MA-14         | M-A           | M-G                    | II         | Dar Bachir (11)                          |
| MA-16         | M-A           | M-G                    | II         | Dar Bachir                               |
| MA-17         | M-A           | M-F                    | III        | Dar Bachir                               |
| MA-18         | M-A           | M-J                    | IV         | Dar Bachir                               |
| MA-20         | M-C           | M-H                    | II         | Dar Bachir                               |

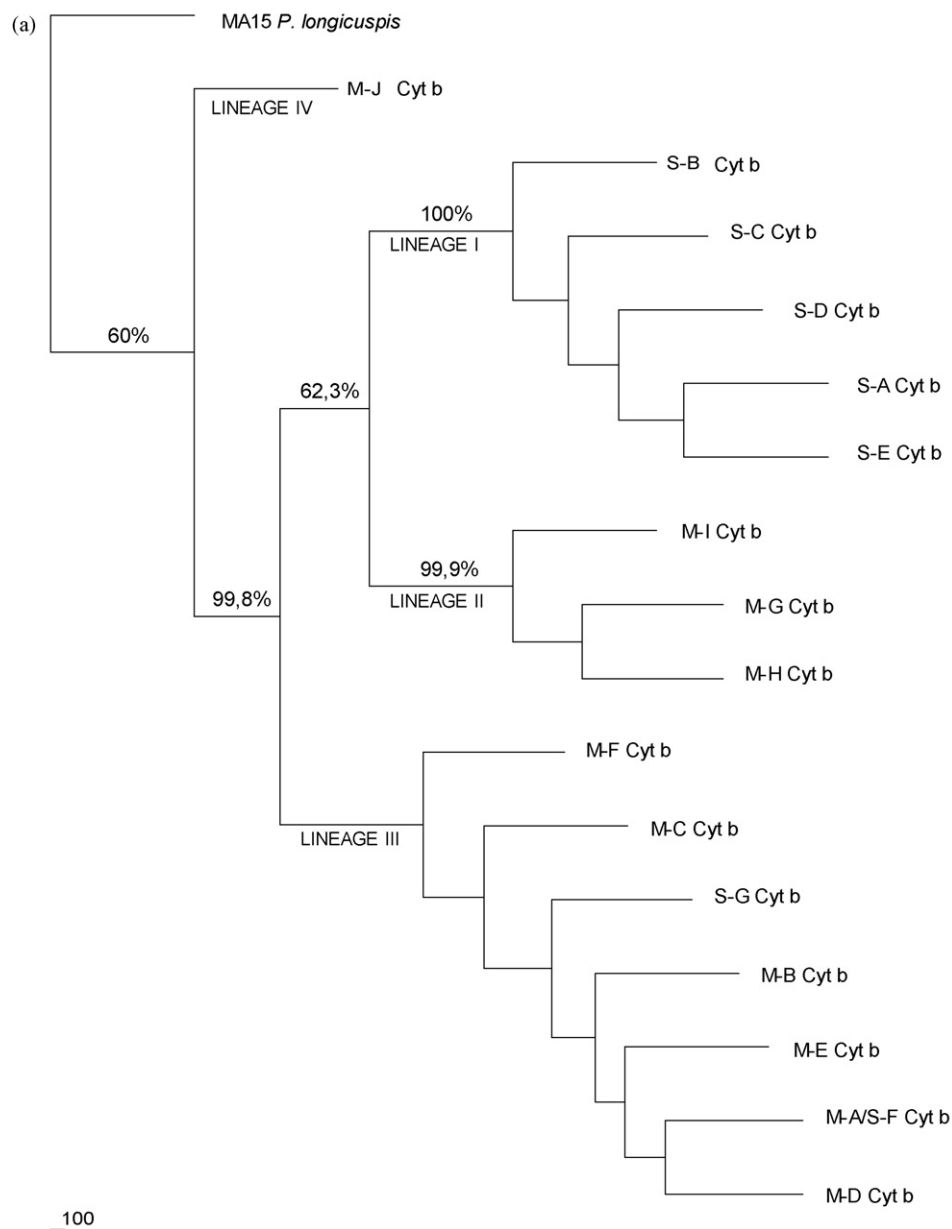
cal range in Spain, being found only in two southern locations (Almáchar and Torvizcón) and in one from the northeast (Sant Just Desvern).

All the Spanish populations studied in the present work in which both markers have been studied shared some haplotypes (ITS or Cyt *b* or both) with the specimens coming from Moroccan populations and two of the Spanish populations (Almáchar and Torvizcón) situated in the south of the country shared the same combination of ITS–Cyt *b* as the Moroccan from Taounate (Table 5). Conversely, the Moroccan population from Dar Bachir does not share any of the haplotypes present in Spain nor from the other Moroccan localities analysed by us. This location is situated in the south of Morocco far from the other two.

In Morocco, cutaneous leishmaniasis due to *L. tropica* is endemic in the south and emergent with epidemical character in several

points of the north and the centre (Guessous-Idrissi et al., 1997). Yahia et al. (2004) identify in Morocco 3 mt DNA lineages of *P. sergenti*. These lineages, and some of the haplotypes, are similar to the ones identified by us in Morocco. Our lineage III coincides with the so-called “Taza lineage” of Yahia et al. (2004) that is found in all the three Moroccan foci of CL researched by the authors. Lineage I is an exclusively Spanish lineage and for this reason it does not have any similarities with the ones found by Yahia et al. (2004).

The presence in Spain of a mitochondrial lineage of *P. sergenti* typically Spanish together with another one common in Morocco where the ACL is endemic and/or emergent could have important epidemiological consequences. It will be necessary to find out if there are or are not differences in the vectorial capacity between these two lineages of *P. sergenti*.



**Fig. 3.** Phylogenetic trees based on mtDNA Cyt *b* sequence data and obtained by distance analysis—F84 model with a UPGMA method of clustering (a) and maximum parsimony method (b). *Phlebotomus longicuspis* is used as an outgroup. The numbers above the branches are bootstrap percentages (1000 replications) for clades supported above the 50%.

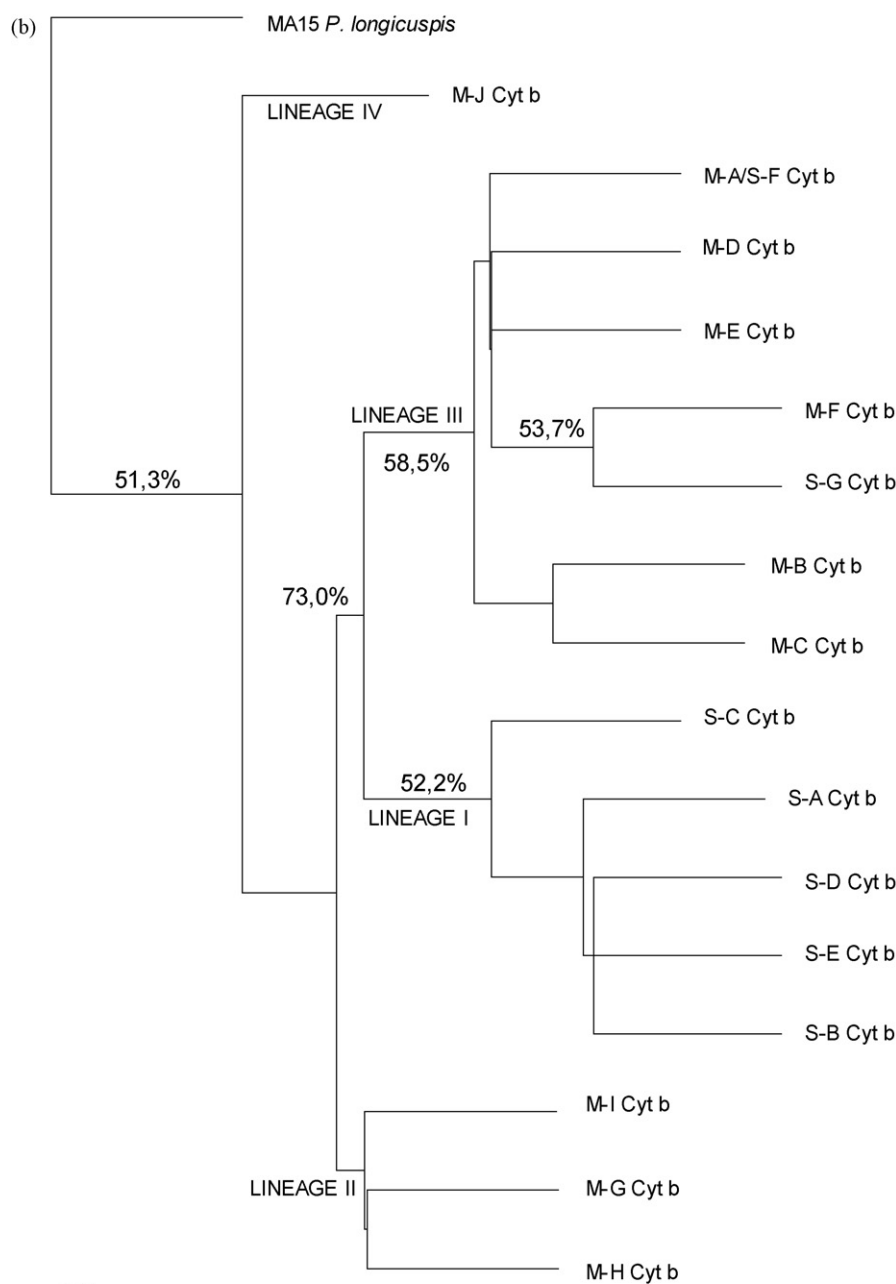


Fig. 3. (Continued).

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